

Predict Before You Explore: Predictive Planning with Specialized Memory for Embodied Question Answering

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Project page: <https://github.com/yuanrr/Pred-EQA>

Abstract

*Embodied Question Answering (EQA) requires agents to navigate 3D environments, accumulate visual evidence, and reason over partial observations to answer questions. However, current agents struggle to maintain coherent, long-horizon behavior: planning remains reactive, causing inconsistent actions, while monolithic memories entangle all observations, hindering retrieval of the sparse but crucial evidence. We address these issues by reframing EQA through the lens of predictive processing, in which coherent behavior emerges from a prediction-correction loop grounded in stable priors. Guided by this perspective, we propose **Predict Before You Explore (Pred-EQA)**, an architecture that integrates predictive planning with specialized memory. A high-level planner predicts where question-relevant evidence is likely to appear and generates a compact set of actionable exploration branches encoding long-horizon intent. A low-level executor then reduces uncertainty within these branches, revising predictions when they fail. A dual-memory system complements this process by separating slowly evolving structural priors from compact, question-relevant visual evidence, enabling consistent planning and efficient evidence accumulation. Through this prediction-guided exploration, *Pred-EQA* achieves coherent trajectories under partial observability. Experiments on *OpenEQA* and *Express-Bench* show that *Pred-EQA* achieves state-of-the-art results in both accuracy and exploration efficiency, demonstrating the benefits of prediction-driven embodied reasoning.*

1. Introduction

Embodied Question Answering (EQA) requires an agent to navigate 3D environments, gather visual evidence, and reason over accumulated observations to answer questions [19]. Unlike object navigation [18], or static visual ques-

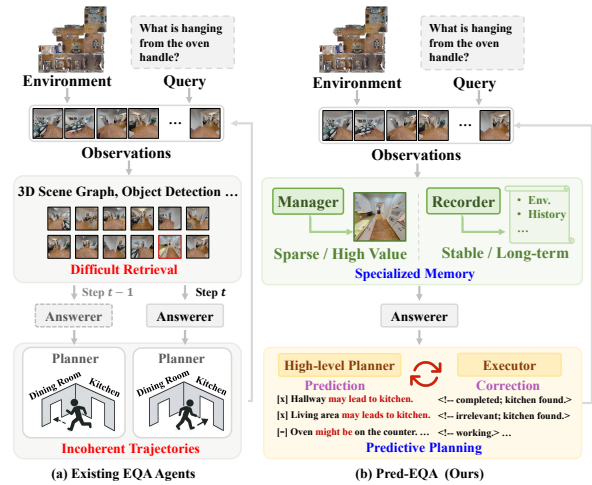


Figure 1. Reactive agents depend on immediate views and monolithic memory, causing inconsistent actions and weak evidence retrieval. *Pred-EQA* introduces predictive planning and structured memory for coherent exploration and reliable reasoning.

tion answering [33, 34], EQA demands both spatial navigation and active scene understanding. While recent Vision-Language Models (VLMs) excel at understanding static scenes [2, 7], extending them to embodied settings is non-trivial because agents must perform sequential planning and maintain persistent memory over long-horizon trajectories.

Recent advances in embodied planning and memory largely come from tool-use or robotic-control tasks, where structured plans or discrete memories support long-horizon actions [3, 6, 39]. These methods work well in relatively static, near-fully observable environments, where an initial view reveals most task-relevant information, enabling plans to be formed in advance and executed with minimal correction. However, such methods are hard to fit in EQA, where agents operate in dynamic, partially observed 3D scenes. Each step reveals only a limited view, and the environment continuously changes as the agent moves, hindering long-horizon planning and consistent memory maintenance. To

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address these challenges, recent EQA studies employ VLM-based embodied agents for semantic planning [4, 24, 29] and visual memory reasoning [13, 26, 30]. For planning, VLMs are often used to reason the next goal based on the current observation and question, but these planners remain largely reactive [13, 30], frequently leading to inconsistent actions across steps and incoherent trajectories. For memory, existing monolithic memory designs [30, 35] face an inherent trade-off. They store extensive visual observations to aid navigation, while most observations are irrelevant to the question, making retrieval of sparse yet crucial evidence difficult. In sum, EQA still suffers from incoherent reactive planning and structural memory conflicts that current VLM-based agents fail to resolve, as in Fig. 1.

To address these issues, we view EQA through the lens of predictive processing [21]. Humans naturally leverage predictive processing to achieve long-term, coherent behavior. Rather than passively reacting to each sensory input, we actively predict future observations based on prior knowledge and goals, and correct our internal prior when predictions fail [10]. This prediction-correction loop provides a structured plan that allows us to maintain coherent intentions under limited or noisy views. It naturally aligns with the demands of EQA. Predictions provide coherent long-horizon objectives that guide exploration toward semantically relevant regions, while feedback refines these hypotheses as the agent moves. Moreover, prediction naturally separates observations into question-agnostic priors and question-relevant evidence. The question-agnostic prior captures stable spatial layouts, serving as a persistent anchor for coherent long-horizon prediction, and can be stored compactly as textual memory. Meanwhile, only sparse, question-relevant evidence is retained as visual memory. This separation enables prediction-driven coherent trajectories and lightweight, task-focused evidence accumulation throughout exploration. Viewed this way, predictive processing provides a foundation for planning and memory architectures that continually propose, test, and revise predictions, enabling coherent trajectories and trustworthy evidence accumulation throughout exploration.

Motivated by this predictive-processing perspective, we introduce **Predict Before You Explore (Pred-EQA)**, an EQA architecture that implements a prediction-correction loop through jointly designed planning and memory mechanisms. On the planning side, we build a high-level planner that first makes high-level predictions about where useful evidence for the question might appear. These predictions define a few actionable exploration branches that encode long-horizon intent. A low-level executor then acts to minimize uncertainty within each branch, trimming branches if a prediction error happens. This loop of planner and executor is significantly different from the reactive behavior and can maintain long-term consistency. On the memory side, we

design a specialized memory system with two clear roles. One records the stable structure of the environment and trajectories, considered as the stable prior crucial for predictive planning. The other keeps only the short-term visual pieces that matter for answering the question. This separation maintains coherent priors for navigation while ensuring efficient, task-focused evidence accumulation.

Our contributions are summarized as follows:

- We present Pred-EQA, a novel EQA framework inspired by predictive processing, establishing that prediction-guided control effectively mitigates partial observability and enables coherent long-horizon reasoning.
- We propose a predictive architecture with 1) hierarchical planners for predicting and exploring actionable branches, and 2) a specialized memory system that separates stable priors from question-related visual evidence.
- We instantiate Pred-EQA within a multi-agent framework. Experiments on OpenEQA [19] and Express-Bench [13] show substantial gains over state-of-the-art methods, including reduced exploration redundancy and significantly higher answer accuracy.

2. Related Works

2.1. Embodied question answering

EQA [8, 19] requires agents to actively explore 3D environments, gather visual evidence, and reason over partial observations. Unlike navigation [18, 37] or static VQA [31, 32], EQA couples perception with long-horizon decision making, posing stronger demands on planning and memory. With advances in VLMs, recent systems leverage VLM-based perception for object detection, semantic labeling, and exploration scoring [4, 24, 26]. However, most remain reactive, relying heavily on current observations without explicit cross-step planning or memory, resulting in inefficient trajectories. To improve planning, several works introduce structured or map-based exploration. GraphEQA [29] uses scene-graph for planning, and frontier or goal-driven strategies exploit VLM semantics for exploration [4, 13, 24]. Despite improvements, these methods still form plans only after new observations. Without predictive intent, agents drift across goals under partial observability. Memory design faces a similar issue: scene graphs, semantic maps, and trajectory encoders [4, 19, 24, 30] fuse dense spatial data with sparse semantic cues, causing retrieval interference and weakening multi-image reasoning.

We interpret these issues through the lens of predictive processing and introduce a prediction-correction loop: a high-level planner predicts promising branches, a low-level executor minimizes uncertainty within predicted branches, and specialized memories stabilize priors while retaining key semantic evidence. This framework enables coherent reasoning and efficient evidence accumulation.

2.2. Multi-Step Planning for LLM Agents

LLM-based planners have shown impressive capabilities in tool-use workflows [6], web navigation [39], and robotic control [3]. These tasks typically assume relatively static and mostly observable environments. Once a plan is generated, the environment rarely deviates. Such settings favor structured workflows over adaptive decision making. In contrast, EQA introduces partial observability, environmental variability, and semantic uncertainty. An LLM planner may generate plausible but incorrect high-level plans based on incomplete information. Strictly following such predefined plans leads to cascading errors. Besides, current LLM planners in EQA remain largely reactive, whose decisions are based on the current observation and immediate feedback [13, 30]. This reactivity leads to short-sighted behaviors and inconsistent trajectories, as plans are repeatedly reformulated without maintaining coherent intent.

Effective embodied planning requires predictive-driven control. Agents must generate hypotheses about promising regions, act to reduce uncertainty, and refine predictions through feedback. Our hierarchical planner follows this paradigm: a high-level planner predicts task-relevant branches, while a low-level executor minimizes local uncertainty. This predictive-corrective loop enables exploration that is both globally coherent and locally adaptive.

2.3. Memory Systems for LLM Agents

Memory plays a central role in long-horizon reasoning for LLM-based agents [17, 20]. Existing frameworks store experiences as sequential logs [9, 27, 38], hierarchical structures, or graph knowledge bases [5, 23, 25, 28]. Inspired by cognitive models, recent works integrate episodic, semantic, and working memory for richer reasoning [14, 16]. While effective in static settings, these designs struggle in embodied tasks, where observations are partial, dynamic, and spatially grounded. EQA demands both a stable global understanding of environments and fine-grained semantic evidence for question answering. Existing memory representations, such as scene graphs [29], semantic maps [4, 24], and video-memory encoders [19], store all observations in a monolithic structure, causing dense trajectory records to overwhelm sparse but crucial evidence. Even filtered memory designs [13, 30, 35] entangle spatial and semantic features, leading to retrieval interference and inconsistent reasoning.

Following the lens of predictive processing, we propose a dual-memory architecture: a long-term structural memory encoding stable priors for coherent planning, and a short-term evidence memory retaining question-relevant observations. This separation anchors predictions to persistent priors while enabling flexible predictions update, achieving both stable navigation and efficient evidence retrieval.

3. Method

We introduce **Predict Before You Explore (Pred-EQA)**, a predictive EQA architecture that transforms reactive navigation into prediction-guided exploration. We begin by motivating our design from predictive processing theory, then describe a minimal multi-agent baseline used to ground our investigation, and finally present the full Pred-EQA architecture, a predictive hierarchical planner equipped with functionally specialized memory, for coherent long-horizon reasoning under partial observability, as in Fig. 3.

3.1. From Reactive to Predictive Processing

EQA requires the agent to navigate under partial observability, where only a fraction of the 3D scene is visible and question-relevant evidence is sparse. We design the agent’s objective in such settings inspired by the variational free-energy bound following active inference formulation [21]:

$$\mathcal{F}(s, q, a) = D_{\text{KL}}[Q(\psi | s, q, a) \| P(\psi, s | q)] - \mathbb{E}_Q[\log P(a | \psi, s, q)]. \quad (1)$$

Here, $Q(\psi)$ is the agent’s belief over latent semantic structure ψ given past observations s , question q , and planned actions a . The KL divergence regularizes the agent’s internal belief to stay coherent with a task-conditioned prior over spatial layout and semantic regularities. The expected log-likelihood favors actions that reduce uncertainty, driving exploration toward locations where prediction errors would be most informative.

Minimizing $\mathcal{F}$ yields a prediction-correction loop: the agent predicts unseen views, acts to validate these predictions, and updates its belief when discrepancies arise. This perspective motivates two design principles in Pred-EQA:

Predictive Hierarchical Planning. The KL term encourages stable predictions of unseen structure, enabling the agent to form long-horizon exploration hypotheses. Pred-EQA implements this via predictive hierarchical planning: a high-level planner proposes coherent exploration intents, while a low-level executor performs uncertainty-reducing actions that refine or correct the predictions.

Functional Specialized Memory. Prediction evaluation requires persistent structural priors, while efficient answering requires compact visual evidence. Pred-EQA adopts a dual-memory scheme: a textual structural memory encoding semantic-spatial regularities, and a visual evidence memory storing only question-relevant observations.

3.2. Multi-Agent Baseline

To systematically study predictive processing in EQA, we build Pred-EQA upon a simple but fully general multi-agent baseline using only RGB-D egocentric input and TSDF-based frontier extraction [24]. The baseline avoids detectors, scene graphs, or semantic maps, ensuring reasoning

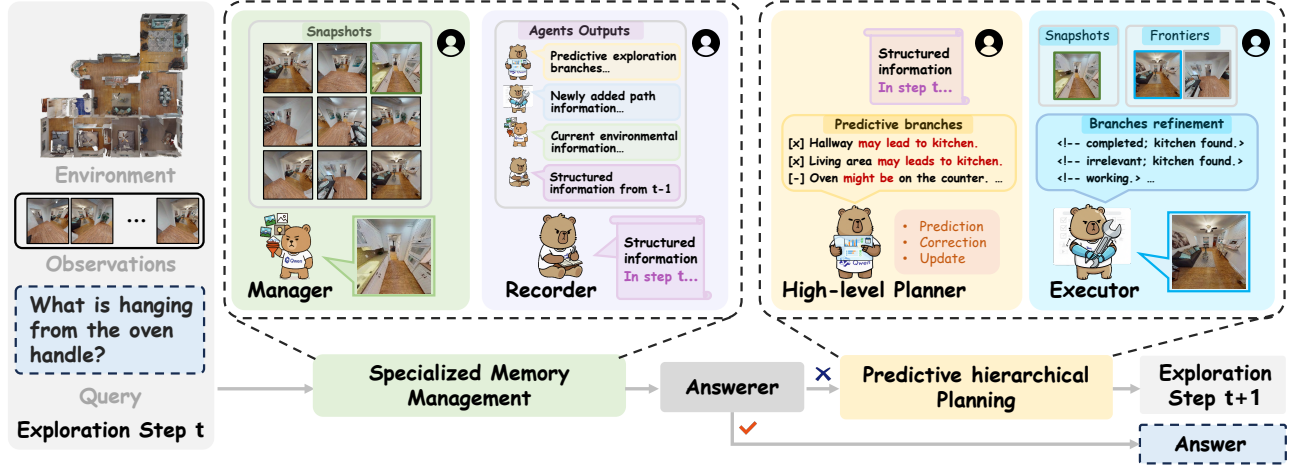


Figure 2. Predict Before You Explore (Pred-EQA), a predictive EQA architecture with predictive hierarchical planning and specialized memory management.

emerges purely from interaction and accumulated observations. At each step t , the agent maintains:

- current egocentric RGB-D images $\mathcal{I}_t^{\text{ego}}$,
- a buffer of previous snapshots $\mathcal{S}_t = \{S_i\}_{i \leq t}$ [30],
- frontier candidates $\mathcal{F}_t$ obtained by clustering TSDF boundary voxels.

There are two agents that operate jointly:

Answerer. Given the question q and snapshot buffer $\mathcal{S}_t$:

$$a_t \leftarrow \text{Answerer}(q, \mathcal{S}_t) \quad (2)$$

It outputs the answer if the visual evidence is sufficient. Otherwise, it emits `Continue Exploration`.

Planner. When further exploration is needed, the planner evaluates frontier relevance:

$$d_t \leftarrow \text{Planner}(q, \mathcal{S}_t, \mathcal{F}_t) \quad (3)$$

The agent navigates toward d_t , gathers a new egocentric observation $\mathcal{I}_{t+1}^{\text{ego}}$, and updates the snapshot buffer:

$$\mathcal{S}_{t+1} = \mathcal{S}_t \cup \mathcal{I}_{t+1}^{\text{ego}} \quad (4)$$

This baseline is fully functional but fundamentally reactive. Pred-EQA replaces this behavior with prediction-guided exploration.

3.3. System Overview: Pred-EQA

Pred-EQA realizes the predictive processing view by structuring EQA as a recurrent prediction-correction-update loop. Unlike reactive frontier-based methods that repeatedly expand visual memory and locally select the next frontier, Pred-EQA maintains explicit predictions about where question-relevant evidence is likely to appear and uses them to guide long-horizon exploration.

At each step t , the system performs three looped operations:

Prediction. The high-level planner predicts semantic exploration branches likely to contain missing evidence, forming predictive exploration branches $\mathcal{T}_t$. These branches encode long-horizon predictive intent, rather than reactive responses to the current view.

Correction. The executor actively tests each branch, verifies or falsifies predictions, and prunes branches as inconsistencies emerge. Exploration thus becomes prediction-driven rather than greedy frontier walking.

Update. The agent then updates two specialized memory modules: Textual memory stores stable semantic and spatial priors, representing the agent’s belief about persistent environment structure. Visual evidence memory stores only question-relevant visual observations. These updated memories serve as refined priors for the next prediction phase, completing the predictive loop.

Through this recurrent loop, Pred-EQA converts exploration from local, reactive frontier-following into global, predictive reasoning, ensuring coherent trajectories, compact memory, and trustworthy evidence acquisition across long-horizon navigation.

3.4. Predictive Hierarchical Planning

Reactive frontier-following treats every step as an isolated decision, making the agent vulnerable to partial observability and goal drift. Pred-EQA instead introduces a high-level predictive planner that forms long-horizon semantic hypotheses, and a low-level executor that actively tests and refines these predictions through interaction with the environment.

High-Level Predictive Planner. At time step t , the high-level planner receives the question q and textual structural memory $\mathcal{H}_t = h_1, \dots, h_{t-1}$ (Sec. 3.5), and predicts a structured branch of semantic hypotheses. Its role is not to pick an immediate frontier, but to predict what information is still missing and where such evidence is likely to appear,

forming a semantic exploration branch. We represent this prediction as a structured predictive branches:

$$\mathcal{T}_t = \{(\tau_i, \sigma_i, \alpha_i)\}_{i=1}^M, \quad (3)$$

where τ_i denotes a hypothesized evidence source or semantic predictions, *e.g.*, “The kitchen might be at the end of this hallway.”, $\sigma_i \in \{\text{pending, active, done}\}$ tracks its progress, and α_i provides a refinement annotation that updates, confirms, or archive predictions inherited from $\mathcal{T}_{t-1}$, *e.g.*, “<!-- irrelevant; kitchen is already identified -->”

These annotations allow the planner to correct previous predictions when new evidence contradicts earlier hypotheses, ensuring consistency across long-horizon reasoning.

A VLM-based predictor produces the updated predictive branches by integrating prior hypotheses with new observations:

$$\mathcal{T}_t \leftarrow \text{H.L.Planner}(q, \mathcal{H}_t, \mathcal{T}_{t-1}). \quad (4)$$

Thus, the high-level planner performs recursive prediction: maintaining, revising, and extending exploration hypotheses as the agent moves through the environment.

Low-Level Executor. Given $\mathcal{T}_t$, curated snapshots $\mathcal{S}'_t$, textual prior $\mathcal{H}_t$ (Sec. 3.5), and frontier set $\mathcal{F}_t$, the executor selects the action aiming to minimize expected free energy at the next step:

$$d_t \leftarrow \text{Executor}(q, \mathcal{T}_t, \mathcal{S}'_t, \mathcal{F}_t, \mathcal{H}_t). \quad (5)$$

Action selection follows three core principles: 1) Maximize information gains: choose frontiers that reduce most uncertainty in predictive branches, 2) Commonsense as guidance: use typical room-object associations to rank frontiers only when no visual clues exist, and 3) Consistency-driven: follow the high-level plan to maintain coherent long-horizon trajectories. These principles implement a local prediction-testing cycle aligned with the high-level hypotheses encoded in $\mathcal{T}_t$.

3.5. Functionally Specialized Memory

Existing EQA memories indiscriminately accumulate raw observations, mixing question-irrelevant frames with sparse but crucial evidence. In contrast, Pred-EQA follows the predictive processing perspective, where memory serves two complementary functions: 1) maintain a question-agnostic prior that anchors long-horizon prediction, and 2) retain question-relevant evidence that informs answer reasoning. To realize this, Pred-EQA implements two functionally specialized memories: a Textual Structural Memory for stable priors, and a Visual Evidence Memory for selective, question-conditioned retention.

Textual Structural Memory. The textual structural memory captures semantic regularities and spatial layout that remain stable over time. Rather than storing raw frames,

the agent converts each verified observation into a symbolic step entry:

$$e_t = (\text{Step}_t, \text{Agent Type}_t, \text{Agent Content}_t, \text{Position}_t), \quad (6)$$

We prompt every agent that needs to receive the snapshot and frontier to first analyze the current state of the environment and its differential comparison with prior knowledge. Therefore, each element records grounded information about the environment. To support predictive planning, these entries are summarized into a textual prior:

$$h_t \leftarrow \text{Recorder}(\{e_{t,i}\}, q), \quad (7)$$

and accumulated as $\mathcal{H}_t = \{h_1, \dots, h_{t-1}\}$. This provides stable priors for future predictions, preventing goal drift.

Visual Evidence Memory. In predictive-processing terms, visual memory should store only the error-signaling evidence needed to validate or refute predictions. Thus Pred-EQA maintains a visual memory that is not a buffer of all past frames, but a sparse, question-relevant evidence set. At each step, a VLM-based manager filters raw snapshot candidates into question-relevant subsets:

$$\mathcal{S}'_t \leftarrow \text{Manager}(q, \mathcal{S}_t, \mathcal{H}_t) \quad (8)$$

A snapshot is kept only when it contributes meaningful evidence: 1) it directly depicts information relevant to the answer, 2) it offers visual-spatial cues that help reduce predictive uncertainty but cannot be captured in textual memory, or 3) it provides a non-redundant view of regions whose state remains uncertain. Snapshots whose content is already covered in $\mathcal{H}_t$ or irrelevant to the question are discarded. For frontier pruning, directions are removed only when both visited and clearly unrelated to the question, ensuring potentially useful within branches.

4. Experiments

4.1. Settings

Simulation and datasets. We conducted evaluations on 2 active EQA benchmarks, including A-EQA, which is a subset of Open-EQA [19], and Express-Bench [13]. Both are based on the Habitat-Matterport 3D (HM3D), which provides hundreds of photo-realistic, diverse indoor 3D scans [22]. See Appendix for more details.

Evaluation Metrics. For quantitative evaluation, A-EQA employs LLM-match score and score averaged by exploration length *i.e.*, LLM-SPL using GPT-4, quantifying answer accuracy and exploration efficiency, respectively. Express-Bench conducts evaluation with GPT-4o-mini using four metrics: C , C^* , E_{path} , and d_T . Among them, C^* reflects answer accuracy, while C and E_{path} jointly capture accuracy and exploration efficiency considering visual evidence consistency. d_T denotes the mean geodesic distance. See Appendix for computing the metrics.

Method	VLM Usage	LLM-Match $\uparrow$	LLM-SPL $\uparrow$
Human Agent	-	85.1	N/A
<i>Blind LLMs</i>			
GPT-4	-	35.5	N/A
GPT-4o	-	35.9	N/A
<i>Agent with Proprietary VLMs</i>			
Memory-EQA [35]	GPT-4o	36.8	-
Explore-EQA [24]	GPT-4o	46.9	23.4
ConceptGraphs [11]	GPT-4o	47.2	33.3
Fine-EQA [13]	GPT-4o-mini	43.3	29.2
MTU3D [40]	GPT-4o	51.1	42.6
3D-Mem [30]	GPT-4o	52.6	42.0
<i>Agent with Open-Source VLMs</i>			
Explore-EQA* [24]	Qwen2-VL 7B	27.4	-
Memory-EQA* [35]	Qwen2-VL 7B	34.6	-
Graph-EQA* [29]	Qwen2-VL 7B	30.1	-
ToolEQA* [36]	Qwen2.5-VL-7B [†]	35.5	-
Pred-EQA	Qwen2.5-VL 7B	46.2	37.8
Pred-EQA	Qwen3-VL 8B	53.3	48.5

Table 1. **Experiments on A-EQA.** Methods with * are reported from Memory-EQA [35] and verified from [36]. [†] means the VLM has been finetuned in 3D data.

Implementation Details We implement Pred-EQA as a pure VLM pipeline with a frontier-based exploration strategy, avoiding scene graph construction or object detection modules. For multi-agent settings, we deploy Qwen3-VL [1] locally using vLLM [15], which enables us to invoke the model with latency and usage patterns similar to GPT-4o API. Each agent is assigned an independent and carefully managed prompt context when invoking the locally deployed model [12]. Following [30], each agent captures 6 egocentric views covering 360° at initialization and then 3 egocentric views covering 120° aligned with its current heading direction at each step. The area within 1.7 meters of the agent’s trajectory is defined as explored region. The frontier is obtained at its boundary following [24]. We limit each episode to a maximum of 50 steps. See Appendix for more detailed settings.

4.2. Comparative Evaluation

Comparison with SOTA Methods. We evaluate Pred-EQA on A-EQA and EXPRESS-Bench to assess answer accuracy, exploration quality, and visual evidence quality. Across both datasets, Pred-EQA consistently establishes a new SOTA, outperforming proprietary and open-source VLM agents. A notable finding is that even with small open-source VLMs, Pred-EQA surpasses strong proprietary agents, highlighting that the performance gains stem from predictive architecture rather than model scale.

On the **A-EQA** dataset in Table 1, Pred-EQA achieves the best performance in both LLM-Match and LLM-SPL metrics. For answer accuracy, with Qwen2.5VL-7B, Pred-EQA outperforms previous open-source VLM agents by

Method	VLM Usage	$C\uparrow$	$C^*\uparrow$	$E_{path}\uparrow$	$d_T\downarrow$
Human Agent	-	-	83.99	-	-
<i>Blind LLMs</i>					
DeepSeek-V3	-	-	59.15	-	-
GPT-4	-	-	58.96	-	-
LLaMA-3-8B	-	-	57.25	-	-
<i>Agent with Proprietary VLMs</i>					
Random Exp. [13]	GPT-4o-mini	36.95	62.75	12.06	7.26
Frontier Exp.[13]	GPT-4o-mini	38.60	60.24	14.55	6.64
Goal Exp. [13]	GPT-4o-mini	38.54	63.34	12.74	6.46
Fine-EQA [13]	Qwen-VL-Plus	40.02	63.06	17.57	6.41
Fine-EQA [13]	GPT-4o-mini	40.55	63.95	16.22	6.43
<i>Agent with Open-Source VLMs</i>					
Fine-EQA [13]	Janus-Pro 7B	39.88	63.35	20.86	6.18
ToolEQA [36]	Qwen2.5-VL 7B [†]	42.21	65.77	25.82	5.25
Pred-EQA	Qwen2.5-VL 7B	47.44	68.54	34.31	5.80
Pred-EQA	Qwen3-VL 8B	52.58	70.54	47.66	5.64

Table 2. **Experiments on Express-Bench.** [†] means the VLM has been finetuned in 3D data. Exp. denotes exploration method.

+10.7%. Scaling to Qwen3VL-8B further surpasses all proprietary VLM agents (+0.7%). In addition, Pred-EQA achieves the most efficient exploration among proprietary counterparts, improving exploration quality by +6.5%. These results indicate that prediction-guided exploration avoids redundant trajectories compared with reactive next-step planners and monolithic-memory agents.

On the **Express-Bench** dataset in Table 2, Pred-EQA again sets a new SOTA in both general and scene-aware correctness, outperforming the SOTA method, *i.e.*, ToolEQA, by +5.23% and +2.77%, respectively. The results show Pred-EQA allows more relevant visual evidence collection, thus improving both general and scene-grounded answer correctness. The gains are even more pronounced in exploration efficiency: Pred-EQA improves the E_{path} metric by over 20% relative to the SOTA method, indicating more efficient and semantically aligned exploration. It is worth noting that Pred-EQA is purely VLM-driven without object detectors [13] or 3D detection tools [36]. Our agent terminates exploration once sufficient evidence is collected rather than physically approaching detected objects. This explains why d_T is not the smallest.

4.3. Ablation Study

Effectiveness of Each Component. Table 3 ablates our predictive planner and specialized memory on A-EQA and Express-Bench using Qwen2.5-VL 7B and Qwen3-VL 8B, demonstrating consistent, model-agnostic effectiveness. The predictive planner alone markedly improves long-horizon reasoning, improving +3.5%/+4.9% LLM-Match and +8.0%/+5.6% LLM-SPL scores on Qwen2.5-VL and Qwen3-VL, respectively. Improvements in E_{path} and C^* also indicate that the planner supports coherent long-

Pred. Plan	Spec. Mem.	A-EQA		Express-Bench	
		LLM-Match↑	LLM-SPL↑	C^* ↑	E_{path} ↑
Qwen2.5-VL 7B		40.8	24.6	41.2	23.8
✓		44.3	32.6	44.7	30.5
	✓	42.3	28.7	43.0	26.8
✓	✓	46.2	37.8	47.4	34.3
Qwen3-VL 8B		45.7	39.5	46.8	35.9
✓		50.8	45.1	49.9	43.4
	✓	47.8	42.2	48.7	39.3
✓	✓	53.3	48.5	52.6	47.7

Table 3. **Effectiveness of Each Component**, predictive hierarchical planner and specialized memory, across A-EQA and Express-Bench datasets.

Planning Method	Spec. Mem.	LLM-Match $\uparrow$	LLM-SPL $\uparrow$
Baseline (Qwen3-VL)	-	45.7	39.5
w. Executor*	-	44.0	39.1
w. TODO list plan $\dagger$	-	47.7	39.9
w. Predictive plan $\dagger$	-	50.8	45.1
Baseline (Qwen3-VL)	✓	47.8	42.2
w. Executor*	✓	48.3	42.8
w. TODO list plan $\dagger$	✓	50.4	43.9
w. Predictive plan $\dagger$	✓	53.3	48.5

Table 4. **Effectiveness of predictive planning**. * means that we prompt the Executor to focus on long-term planning, but it is fundamentally reactive. $\dagger$ means different long-term planning settings with a default Executor.

horizon exploration. Using only the specialized memory also boosts accuracy and navigation efficiency for +1.5% LLM-Match and +4.1% LLM-SPL on Qwen2.5-VL by ensuring cleaner retrieval over monolithic designs. Combining both modules yields the best gains. Under Qwen2.5-VL, LLM-Match improves by +6.0% and LLM-SPL by +13.2%. Qwen3-VL shows +7.6% and +9.0% gains, respectively. Express-Bench metrics follow the same pattern. These additive improvements highlight the synergy: predictive planning supplies structured hypotheses, while specialized memory preserves critical evidence for validation.

Effectiveness of Predictive Planning Table 4 compares different planning strategies with and without specialized memory. Without specialized memory, predictive planner improves LLM-Match and LLM-SPL by +5.1% and +5.6%, respectively, outperforming both the reactive Executor and the heuristic TODO-list planner. This confirms that in partially observed environments, simply prompting reactive agents or relying on flat subgoal lists cannot produce reliable long-horizon behavior. TODO-list planning commits to a fixed set of subgoals that quickly become outdated in partially observed 3D environments, leading to brittle trajectories. In contrast, predictive planning generates coherent multi-step hypotheses to reduce redundant exploration. When equipped with specialized memory, the predictive planner achieves the best overall performance, *i.e.*, +5.5%

Method	Pred. Plan	LLM-Match $\uparrow$	LLM-SPL $\uparrow$
Baseline (Qwen3-VL)	-	45.7	39.5
w. 3D-Mem [30]	-	48.8	42.7
w. Manager	-	47.5	41.5
w. Manager*	-	49.2	43.1
w. Spec. Mem.	-	47.8	42.2
Baseline (Qwen3-VL)	✓	50.8	45.1
w. 3D-Mem [30]	✓	51.7	47.5
w. Manager	✓	52.2	47.2
w. Manager*	✓	53.1	48.0
w. Spec. Mem.	✓	53.3	48.5

Table 5. **Effectiveness of specialized memory**. * means that we prompt the Manager to focus on both environment recording and question-relevant evidence maintenance.

VLM Usage	Scale	LLM-Match $\uparrow$	LLM-SPL $\uparrow$
Qwen2.5-VL Series	3B	38.6	32.2
	7B	46.2	37.8
	32B $\dagger$	51.4	48.3
Qwen3-VL Series	4B	50.8	46.2
	8B	53.3	48.5
	30B-A3B $\dagger$	55.0	52.4
	32B $\dagger$	56.3	53.9

Table 6. **Scalability of Pred-EQA on different architectures and sizes**. $\dagger$ means the FP8 quantized version.

LLM-Match and +6.3% LLM-SPL. This synergy arises because predictive plans provide structured trajectories, while specialized memory preserves the evidence necessary to verify and refine these long-horizon predictions.

Effectiveness of Specialized Memory Table 5 evaluates memory designs under both reactive and predictive settings. Manager* maintains a global visual record similar to 3D-Mem, but differs in its filtering mechanism: 3D-Mem removes redundant frames via an object detector, whereas Manager* relies on the VLM’s judgment. The latter yields a slightly better accuracy. Manager and our specialized memory retain only question-relevant frames, resulting in extremely compact memories, *e.g.*, < 3 frames retained each step on average. However, without predictive planning, their gains are smaller than Manager*, as reactive agents lack coherent trajectories to reliably visit question-relevant regions, leaving the compact memory insufficient. With predictive planning enabled, specialized memory achieves the best performance while keeping the smallest memory footprint. This shows that long-horizon structure compensates for the reduced memory size and allows the specialized memory to preserve only the most critical evidence.

Scalability with Model Size. Table 6 investigates the scalability of Pred-EQA. As the model size increases, Pred-EQA continues to scale favorably, with +2~5% in LLM-Match and +4~6% in LLM-SPL. This demonstrates that stronger VLMs can further leverage our predictive planning and specialized memory to achieve more consistent and accurate long-horizon reasoning.

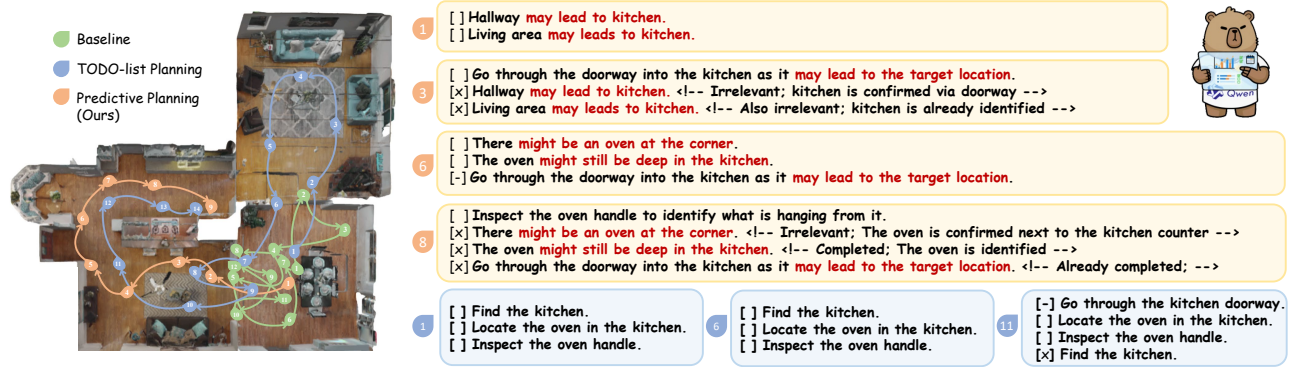


Figure 3. **Visualization of planning behaviors across different agents** including the reactive Baseline, the TODO-list planner, and predictive planning in our Pred-EQA.

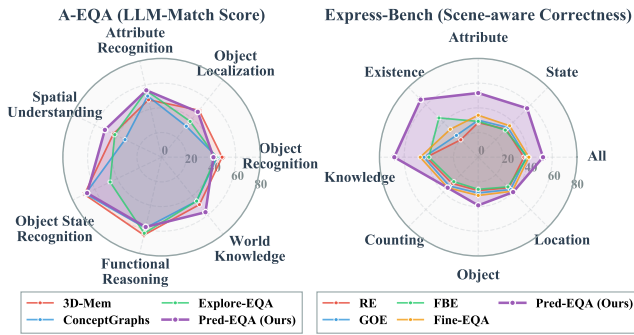


Figure 4. **Visualization Studies on Question Categories** using LLM-Match on A-EQA and scene-aware correctness C on Express-Bench.

Visualization Studies on Predictive Planning. Fig. 4 compares Pred-EQA with the reactive Baseline and TODO-list planning. The Baseline exhibits a strong step-wise drawback: it greedily moves toward the most visually salient region and frequently performs unnecessary backtracking. For example, when entering the living room, it keeps going back to the dining room because it thought the dining room was closer to the kitchen. The TODO-list planner introduces a persistent list of subgoals, but its planning remains essentially uninformative. For example, it often generates vacuous items such as “find the kitchen”, regardless of current context. In contrast, Pred-EQA constructs predictive exploration branches that encode concrete, testable hypotheses, e.g., the kitchen being behind a doorway or the oven near a specific counter. As the agent moves, the prediction-correction loop validates or eliminates branches, preventing the buildup of obsolete goals. This leads to coherent trajectories that reduce redundant exploration and focus the agent on truly informative regions.

Visualization Studies on Question Categories. Fig. 4 compares Pred-EQA with recent agents across LLM-Match on A-EQA and scene-aware correctness C on Express-Bench. On A-EQA, Pred-EQA consistently outperforms

across nearly all capabilities, particularly in Spatial Understanding and Functional Reasoning, highlighting the benefit of prediction-guided planning. On Express-Bench, Pred-EQA achieves large improvements in Existence, State, and Knowledge reasoning, which require trustworthy evidence accumulation over extended trajectories. These results validate that the prediction-correction loop produces coherent exploration behaviors and enables effective retrieval of sparse, question-relevant observations.

5. Conclusion

In this work, we reinterpret EQA through the lens of predictive processing and show that prediction-guided exploration offers a novel solution to the limitations of reactive planning and monolithic memory designs. Building on this perspective, we introduced Pred-EQA, an architecture that couples hierarchical prediction-correction planning with a structured memory system separating stable priors from sparse, question-centric evidence. This design enables agents to form coherent long-horizon exploration intents, adaptively revise predictions under partial observability, and retain only the information needed for trustworthy reasoning. Experiments on OpenEQA and Express-Bench validate that Pred-EQA yields more consistent trajectories, reduces redundant exploration, and achieves SOTA compared with recent VLM-based embodied agents. We believe this predictive-processing framework opens a promising direction for future embodied agents that demand both long-term coherence and efficient evidence accumulation.

Acknowledgments

This work was supported by the National Natural Science Foundation of China under Grants (No.62325206, 62532003, 62301276), the Key Research and Development Program of Jiangsu Province under Grant BE2023016-4, and the China Postdoctoral Science Foundation 2025M773507.

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